

Symmetry Transmutation from RG Wall Correspondences

Codex

ABSTRACT: An RG wall determines a local kernel of junction spaces between UV and IR backgrounds. We prove that a coherent local representation of this kernel produces a symmetric-monoidal map $\Phi : \mathfrak{B}_{\text{UV}}^{\nabla} \rightarrow \mathfrak{B}_{\text{IR,tr}}^{\nabla}$, unique up to contractible choice. Right-principality of the IR wall action is sufficient. On the invertible core, Φ recovers the classifying-space map of Seiberg and Seifnashri, including the obstruction $BG \rightarrow B^2K$ from a central extension, and flat restriction transports stratified noninvertible networks. If its degree-preserving codimension-one part is a strong unitary pivotal tensor functor, annularization produces a tube-algebra homomorphism into a stable corner. Differential lifts preserve full gauge descent and curvature. Given a local Wilsonian factorization with a pointwise nonzero monoidal intertwiner, the UV anomaly is the pulled-back IR anomaly tensored with the wall contribution. A concrete flow must provide the wall representation and the later differential and Wilsonian data.

KEYWORDS: generalized symmetry, noninvertible symmetry, tube algebra, differential cohomology, anomaly matching, RG interface

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1 Introduction

Seiberg and Seifnashri describe symmetry transmutation by a local assignment

$$B_{\text{IR}} = \Phi(A_{\text{UV}}) \quad (1.1)$$

from UV backgrounds to backgrounds for a symmetry of a different form in the IR [1, Sec. 6.2 and App. B]. For a central extension

$$1 \longrightarrow K \longrightarrow \tilde{G} \longrightarrow G \longrightarrow 1, \quad (1.2)$$

their assignment is the obstruction map $BG \rightarrow B^2K$. Classifying-space maps carry invertible topological backgrounds. Noninvertible defects also carry fusion and junction data, which tube algebras detect [9, Sec. 2.3]. Continuous symmetries couple to nonflat connections [2, Secs. 2–3]. On a stratified spacetime, we place connections on top-dimensional strata and defects or junctions on lower-dimensional strata. Their smooth stack of linear defect higher categories is the geometric input.

The wall-junction spaces form a kernel $\mathcal{K}_R$, contravariant in the UV background and covariant in the IR background. For each $x \in \mathfrak{B}_{\text{UV}}^\nabla$, require the restricted functor

$$\mathcal{K}_R(x, -)|_{\mathfrak{B}_{\text{IR}, \text{tr}}^\nabla}$$

to be locally representable. Choose representing objects with the descent, collar, refinement, and monoidal coherences required below. They assemble into a local symmetric-monoidal functor

$$\Phi : \mathfrak{B}_{\text{UV}}^\nabla \longrightarrow \mathfrak{B}_{\text{IR}, \text{tr}}^\nabla, \quad (1.3)$$

unique up to contractible choice. Right-principality supplies a sufficient interface condition.

Representation gives the same map on the invertible core and on flat stratified backgrounds. Annularization is available for its degree-preserving strong unitary pivotal part. Differential transport keeps descent and curvature, and anomaly transport requires a point-wise nonzero map between determinant lines. The model-dependent work is constructing the wall representation and its differential and Wilsonian extensions.

2 Backgrounds with connections and defects

2.1 The differential defect stack

Connections and defect labels occupy different strata and transform under the same local symmetries and gluing maps. We encode them together by fibering network categories over a smooth stack of connective fields.

Let StratFam_d be the site of smooth families of tangentially-structured, conically smooth stratified manifolds of dimension at most d , with collared strata. Write ManFam_d for its full unstratified subsite. Fix a presentable symmetric-monoidal linear ∞ -category $\mathcal{V}$. Let $\mathcal{W}_d$ be a full subcategory of $\mathcal{V}$ -enriched (∞, d) -categories closed under the limits, colimits, relative tensor products, and idempotent completions used below.

Definition 2.1 (Differential defect background stack). Let

$$\mathcal{C}^\nabla : \mathbf{ManFam}_d^{\mathrm{op}} \longrightarrow \mathbf{Spaces}$$

be a hypercomplete smooth stack of connective fields. Write $\int \mathcal{C}^\nabla$ for its Grothendieck construction over $\mathbf{ManFam}_d$. A differential defect coefficient system is a cartesian fibration

$$p : \mathcal{E}^\nabla \longrightarrow \int \mathcal{C}^\nabla$$

whose fiber over (M, A) , together with an admissible stratification Σ , is a symmetric-monoidal functor

$$\mathcal{D}_{A,\Sigma} : \mathrm{Exit}(M_\Sigma) \longrightarrow \mathcal{W}_d.$$

Its transport along gauge equivalences, pullbacks of smooth families, and collared refinements is part of the coefficient system. Write

$$\mathrm{Net}_A(M_\Sigma) := \mathrm{Sec}_{\mathrm{c,oplax}}(\mathrm{Exit}(M_\Sigma), \mathcal{D}_{A,\Sigma}). \quad (2.1)$$

The differential defect background stack is the homotopy Grothendieck construction

$$\mathfrak{B}_{\mathcal{D},\mathrm{str}}^\nabla(M_\Sigma) := \int_{A \in \mathcal{C}^\nabla(M)}^h \mathrm{Net}_A(M_\Sigma). \quad (2.2)$$

The chosen coefficient system must satisfy the following conditions.

- (D1) It satisfies smooth hyperdescent and sends disjoint unions to completed tensor products.
- (D2) Collar gluing is computed by relative tensor product.
- (D3) A colored stratified-disk operad assigns labels with their incidence, duality, and gluing data. In the fusion-category convention, codimension-one strata carry objects and codimension-two strata carry morphisms. A unitary theory carries a compatible dagger.
- (D4) Defect labels are invariant under constructible isotopy. Full-dimensional connective labels vary smoothly and retain their connection and curvature data.
- (D5) On an unstratified chart, the maximal groupoidal core is a smooth stack of connective fields. Abelian higher-form fields carry complete Čech–Deligne descent. A Lie group G gives the nonabelian stack $\mathbf{B}G_\nabla$ with gauge descent.

The exit-path construction supplies the constructible topological indexing category in (2.1) [11, Secs. 3–4 and Thm. 0.0.1]. The smooth connective stack and its action on the fibers $\mathcal{D}_{A,\Sigma}$ must be specified separately. Top-dimensional strata carry connective fields, walls carry defects, codimension-two strata carry junctions, and deeper strata carry higher coherences.

A morphism of background stacks is *regular-local* when its connection-form component is induced in every smooth chart by a specified finite-jet map and its transition data define a morphism of the full descent stacks. The jet map preserves gauge equivalences, curvature identities, integral periods, and higher gauge coherences.

Let $\text{Ref}(M)$ be the ∞ -category of admissible collared stratifications, with collared refinements and constructible isotopies as morphisms. Let W_u be the class of unit-labeled refinements. The coefficient system supplies the transition functors on the localization $\text{Ref}(M)[W_u^{-1}]$. When the objectwise colimit exists and hypersheafification preserves collar excision, define

$$\mathfrak{B}_{\mathcal{D}}^{\nabla} := a_{\text{hyp}} \text{Kar} \left[M \longmapsto \text{hocolim}_{\Sigma \in \text{Ref}(M)[W_u^{-1}]} \mathfrak{B}_{\mathcal{D}, \text{str}}^{\nabla}(M_{\Sigma}) \right]. \quad (2.3)$$

The operator a_{hyp} denotes hypersheafification. A codimension-changing map includes a coherent morphism between the colored stratified-disk operads of the two coefficient systems. That morphism specifies output refinements and their incidence coherences. The marked localization in (2.3) identifies unit refinements. Its objects include regions carrying both a smooth connection and an embedded noninvertible defect.

2.2 Standard examples

Example 2.2 (Ordinary and higher groups). For a Lie group G , the invertible unstratified sector is the moduli stack $\mathbf{B}G_{\nabla}$ of principal G -bundles with connection. Its value on M is the groupoid $\text{Bun}_G^{\nabla}(M)$. The same construction gives $\mathbf{B}\mathbb{G}_{\nabla}$ for higher groups and gerbes. The free linear and idempotent-complete envelope embeds this groupoid-valued example in Definition 2.1.

Example 2.3 (Fusion category). For a finite pivotal fusion category $\mathcal{C}$, a flat stratified background is a $\mathcal{C}$ -labeled network. Codimension-one strata carry objects of $\mathcal{C}$, and their codimension-two junctions carry morphisms. Refinement is governed by the associator and its pentagon. Direct sums and idempotent completion add condensation defects built from separable idempotents and their modules [4]. Condensable algebras in a modular tensor category give a concrete braided example [5, 6]. Curvature is identically zero in this purely topological example.

For a principal G -connection, the construction in (2.2) takes the form

$$\mathfrak{B}_{\mathcal{D}}^{\nabla}(M) \simeq \int_{(P,A) \in \mathbf{B}G_{\nabla}(M)}^h \text{Net}_{\mathcal{S}(P,A)}(M), \quad (2.4)$$

where $\mathcal{S}$ is a specified smooth pseudofunctor satisfying descent. Its value on a gauge morphism supplies transport between the network fibers. We use $\int^h$ for the homotopy Grothendieck construction.

The product system

$$\mathfrak{B}_{\text{prod}}^{\nabla}(M_{\Sigma}) := \mathfrak{B}_G^{\nabla}(M) \boxtimes \text{Net}_{\mathcal{C}}(M_{\Sigma}), \quad (2.5)$$

has $\mathcal{S}(P, A) = \text{Net}_{\mathcal{C}}$ constant. Descent and gluing leave the connective and defect factors independent. Coupling them requires an action of the connection on defect fibers together with a colored-operad morphism.

2.3 Flat, invertible, and annular operations

Suppress the defect-system subscript $\mathcal{D}$ and define:

1. $\text{Core}(\mathfrak{B}^\nabla)$ is the substack whose defect labels are invertible. The maximal groupoidal substack retains all objects and restricts morphisms to equivalences. If the connective fields have a curvature morphism to a curvature stack $\mathfrak{F}$, with zero section 0, define

$$\text{Core}_{\text{flat}}(\mathfrak{B}^\nabla) := \text{hofib}_0(\text{curv} : \text{Core}(\mathfrak{B}^\nabla) \rightarrow \mathfrak{F}).$$

2. $\text{Flat}(\mathfrak{B}^\nabla)$ is the pullback of the network stack over the zero-curvature connective substack. Its construction uses the action of connective fields on network fibers.
3. $\text{Ann}(\text{Flat } \mathfrak{B}^\nabla)$ is the annular category obtained by evaluation and gluing. When the relevant fiber is a finite semisimple rigid C^* -tensor category with a source-tagged additive generator, its endomorphism algebra is the tube algebra.

A *stratified differential transmutation map* contains a regular-local morphism $\phi : \mathcal{C}_{\text{UV}}^\nabla \rightarrow \mathcal{C}_{\text{IR}}^\nabla$ between connective stacks and a morphism of colored stratified-disk operads

$$\sigma_\Phi : \text{Disk}_{\text{UV}}^{\text{col}} \longrightarrow \text{Disk}_{\text{IR}}^{\text{col}}.$$

The operad map induces compatible refinement functors $\sigma_{\Phi, M} : \text{Ref}_{\text{UV}}(M) \rightarrow \text{Ref}_{\text{IR}}(M)$. On every labeled stratified family it supplies coherent symmetric-monoidal functors

$$\Phi_{M, \Sigma, A} : \text{Net}_A^{\text{UV}}(M_\Sigma) \longrightarrow \text{Net}_{\phi(A)}^{\text{IR}}(M_{\sigma_\Phi(\Sigma)}).$$

Compatibility holds for smooth pullback, active operad composition, marked unit refinements, constructible isotopies, and collar gluing. Descent through (2.3) gives a regular-local functor $\Phi : \mathfrak{B}_{\text{UV}}^\nabla \rightarrow \mathfrak{B}_{\text{IR}}^\nabla$. Applying Core, Flat, and Ann gives its restrictions when Φ preserves the domains of these operations.

3 RG interfaces

The junctions of an RG interface form an enriched profunctor. Representability in the IR variable turns that profunctor into a background map.

An RG interface R is *admissible* if its UV and IR background actions are $\mathcal{V}$ -enriched functors compatible with restriction, disjoint union, and collar excision on a generating family of stratified disks. Admissibility also includes smooth-family hyperdescent and the coend composition law for collared disk gluing. On such a disk D , let $\text{Def}(R; D)$ be the $\mathcal{V}$ -enriched category of wall junctions and write

$$L_D(x) = R \circ \rho_{\text{UV}}(x), \quad J_D(y) = \rho_{\text{IR}}(y) \circ R. \quad (3.1)$$

The diskwise kernel

$$\mathcal{K}_{R,D}(x, y) := \text{Map}_{\text{Def}(R;D)}(L_D x, J_D y) \quad (3.2)$$

is contravariant in x and covariant in y . Admissibility assembles these kernels into a local lax symmetric-monoidal profunctor

$$\mathcal{K}_R : (\mathfrak{B}_{UV}^\nabla)^{\text{op}} \otimes \mathfrak{B}_{IR}^\nabla \longrightarrow \mathcal{V}.$$

The wall-junction mapping objects take values in the linear enrichment $\mathcal{V}$.

Fix an objectwise full, local, idempotent-complete transmitted substack

$$\iota_{\text{tr}} : \mathfrak{B}_{IR, \text{tr}}^\nabla \hookrightarrow \mathfrak{B}_{IR}^\nabla$$

closed under pullback, refinement, collar gluing, tensor product, and the selected idempotent completion. Let

$$\mathcal{K}_R^{\text{tr}} := \mathcal{K}_R|_{(\mathfrak{B}_{UV}^\nabla)^{\text{op}} \otimes \mathfrak{B}_{IR, \text{tr}}^\nabla}.$$

Proposition 3.1 (Coherent enriched representability). *A transmitted background map represented by $\mathcal{K}_R^{\text{tr}}$ is a pair (Φ, η) , where Φ has the required stratified locality and*

$$\eta_{x,y} : \mathcal{K}_R^{\text{tr}}(x, y) \xrightarrow{\simeq} \text{Map}_{\mathfrak{B}_{IR, \text{tr}}^\nabla}(\Phi x, y).$$

The equivalence is natural in both variables and preserves restriction, collar gluing, refinement, and the monoidal structure. Such a pair exists exactly when every $\mathcal{K}_R^{\text{tr}}(x, -)$ is representable and its representing objects admit a coherent local symmetric-monoidal choice. Any such choice defines

$$\Phi : \mathfrak{B}_{UV}^\nabla \longrightarrow \mathfrak{B}_{IR, \text{tr}}^\nabla. \quad (3.3)$$

The resulting functor is unique up to contractible choice.

Proof. The chosen representatives carry the required local coherences. Enriched Yoneda lifts each UV morphism and identifies the compatible lift uniquely. Applying enriched Yoneda to the tensor, unit, restriction, collar, and refinement equivalences gives the corresponding structure maps. Evaluation against every IR object proves their coherence. $\square$

Definition 3.2 (Right-principal transmitted sector). The wall is *right-principal* on the chosen transmitted sector if the local action J is $\mathcal{V}$ -enriched fully faithful and strong symmetric-monoidal, with compatibility under pullback, refinement, colored operad action, and collar gluing. Require L to have the same locality and monoidal properties. For each x , the infinity-groupoid of pairs

$$(y, \theta), \quad \theta : L_D(x) \xrightarrow{\simeq} J_D(y),$$

must form a specified compatible section over the disk and refinement infinity-category.

Theorem 3.3 (Interface transport). *The section in Definition 3.2 determines Φ , up to contractible choice, together with pull-through equivalences*

$$\theta_x : L_D(x) \xrightarrow{\simeq} J_D(\Phi x). \quad (3.4)$$

Proof. The section gives $(\Phi_D x, \theta_x)$. Full faithfulness of J_D makes this pair unique up to contractible choice. For $f : x \rightarrow x'$, define $\Phi_D(f)$ by

$$\theta_{x'} L_D(f) = J_D(\Phi_D f) \theta_x. \quad (3.5)$$

Full faithfulness transfers the tensorator and unit maps from the strong monoidal structures on L and J . Their diagrams commute after applying J_D , and faithfulness brings the equalities back to the source. Compatibility of the section glues the diskwise construction by descent. $\square$

Example 3.4 (A nonfaithful wall). Let both symmetry categories be $\text{Vec}_{\mathbb{Z}_2}$, and take the module wall $R = \text{Vec}$ on which both simple objects act as the identity. The identity and trivial homomorphisms $\mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ therefore induce the same wall action. The kernel is compatible with two distinct background maps.

A central extension gives a weak 2-functor that sends every wall to the unit and retains a nontrivial compositor at junctions. Section 4 constructs this codimension shift. Noninvertible output needs stratified data beyond an invertible pseudofunctor compositor. The colored-operad morphism after (2.3) chooses the output strata and labels on each disk. Compatibility with active maps gives subdivision coherence. Codimension-changing transmutation includes this morphism as input.

We call the process *transmutation* when a transmitted defect acts trivially at its original codimension in the IR and leaves nontrivial junction or contact data at higher codimension. An IR defect outside the image of Φ , or the right support of $\mathcal{K}_R$, is emergent. A projective action on extended objects within one fixed theory is fractionalization [3]. A background ending on the RG wall exhibits symmetry breaking.

4 Invertible and flat backgrounds

4.1 Classifying stacks and flat local systems

When Φ preserves the unstratified invertible topological sector, let $\mathbb{G}_T$, for $T \in \{\text{UV}, \text{IR}\}$, denote the corresponding invertible higher symmetry. Write Core_{top} for the operation that first restricts to invertible defect labels and then forgets connective data. The resulting core is

$$\text{Core}_{\text{top}}(\mathfrak{B}_T^\nabla)(M) \simeq \text{Map}(M, B\mathbb{G}_T). \quad (4.1)$$

By Yoneda, the restriction of Φ is determined by

$$\Phi^{\text{top}} : B\mathbb{G}_{\text{UV}} \longrightarrow B\mathbb{G}_{\text{IR}}. \quad (4.2)$$

A background $A : M \rightarrow B\mathbb{G}_{\text{UV}}$ is sent to $\Phi^{\text{top}} \circ A$. A homomorphism $\varphi : G_{\text{UV}} \rightarrow G_{\text{IR}}$ gives the usual pushforward $\varphi_* A$.

Flatness imposes a separate condition. For a Lie group G ,

$$B\mathcal{G}_{\text{flat}}(M) \simeq \text{Fun}(\Pi_1(M), BG).$$

On connected pointed M , the formula becomes $\text{Hom}(\pi_1 M, G) // G$. For discrete G , this is the stack of topological G -bundles. Preservation of zero curvature is the condition for a differential map to restrict to this substack.

4.2 Central extensions

Let K be abelian and central in $\tilde{G}$. The extension (1.2) determines an obstruction map

$$e : BG \longrightarrow B^2K. \quad (4.3)$$

Set $\Phi_e = e$. For a G -bundle on M , take transition functions g_{ij} on a good cover and choose lifts $\tilde{g}_{ij} : U_{ij} \rightarrow \tilde{G}$. Define

$$b_{ijk} = \tilde{g}_{ij}\tilde{g}_{jk}\tilde{g}_{ik}^{-1} \in K. \quad (4.4)$$

Associativity gives

$$b_{ijk}b_{ikl} = b_{jkl}b_{ijl}. \quad (4.5)$$

If $\tilde{g}'_{ij} = \lambda_{ij}\tilde{g}_{ij}$, then

$$b'_{ijk} = \lambda_{ij}\lambda_{jk}\lambda_{ik}^{-1}b_{ijk}. \quad (4.6)$$

The class $[b] \in H^2(M; K)$ obstructs a $\tilde{G}$ -lift, and

$$B_{\text{IR}} = \Phi_e(A_{\text{UV}}) = [b].$$

For abelian groups this is the connecting homomorphism

$$H^1(M; G) \xrightarrow{\beta_e} H^2(M; K), \quad \beta_e([A]) = [b]. \quad (4.7)$$

For $G = K = \mathbb{Z}_2$ and $\tilde{G} = \mathbb{Z}_4$, as in [1, Sec. 6.2],

$$B_{\text{IR}} = \frac{1}{2}\delta\tilde{A} \pmod{2}. \quad (4.8)$$

For discrete G , choose a normalized section $s : G \rightarrow \tilde{G}$ and set $\gamma(g, h) = s(g)s(h)s(gh)^{-1}$. The cocycle equation

$$\gamma(h, k)\gamma(g, hk) = \gamma(g, h)\gamma(gh, k) \quad (4.9)$$

is the pentagon for a weak 2-functor $\Phi_\gamma : BG \rightarrow B^2K$ with

$$\Phi_\gamma(g) = \text{id}, \quad \mu_{g, h} = \gamma(g, h). \quad (4.10)$$

The functor maps individual G -walls to the unit wall and assigns their fusion junction the K -label $\gamma(g, h)$.

For a character $\chi : K \rightarrow U(1)$,

$$U_g U_h = \chi(\gamma(g, h)) U_{gh}. \quad (4.11)$$

The Čech obstruction and weak 2-functor represent the same extension class, and the projective action is its pushforward along χ . For Lie groups, work on smooth manifolds with open-cover descent. Denote the universal smooth-stack class by

$$e \in H^2(\mathbf{B}G; \underline{K}) \cong \pi_0 \text{Map}_*(\mathbf{B}G, \mathbf{B}^2K).$$

Seiberg and Seifnashri give the classifying-space description in [1, App. B]. Evaluation on a principal bundle gives the lifting-gerbe obstruction [10, Secs. 2–3]. A connective lift requires

a chosen multiplicative differential refinement of this extension, with its curvature and integral descent data.

For discrete coefficients,

$$[BG, B^{p+1}K] \cong H^{p+1}(BG; K) \quad (4.12)$$

and a class in $H^{p+1}(BG; K)$ sends a zero-form G -background to a p -form K -background.

5 Noninvertible defects and tube algebras

Annularization turns degree-preserving defect transport into an operator-algebra map.

5.1 Annular category

Let $\mathcal{C}$ be a finite semisimple rigid C^* -tensor category with a dagger-compatible pivotal structure. Its annular category has the coend presentation

$$\mathrm{Hom}_{\mathrm{Ann}(\mathcal{C})}(j, i) = \int^{a \in \mathcal{C}} \mathrm{Hom}_{\mathcal{C}}(a \otimes j, i \otimes a). \quad (5.1)$$

For simple representatives this is

$$\bigoplus_{a \in \mathrm{Irr}(\mathcal{C})} \mathrm{Hom}_{\mathcal{C}}(a \otimes j, i \otimes a)$$

modulo the dinaturality relation. With the source-tagged additive generator

$$\Omega_{\mathcal{C}} = \bigoplus_{i \in \mathrm{Irr}(\mathcal{C})} i, \quad \mathrm{Tub}(\mathcal{C}) = \mathrm{End}_{\mathrm{Ann}(\mathcal{C})}(\Omega_{\mathcal{C}}), \quad (5.2)$$

annulus gluing defines tube multiplication. After choosing standard solutions of the conjugate equations, reflection defines the dagger [9, Sec. 2.3]. A different choice gives a gauge-isomorphic tube algebra. Lasso operators realize it in two-dimensional CFT [7, Sec. 4.1]. Choi, Rayhaun, and Zheng construct the boundary and interface versions in [8, Secs. 2.2 and 3.2]. Set $\Omega_{\mathrm{UV}} = \Omega_{\mathcal{C}_{\mathrm{UV}}}$ and $\Omega_{\mathrm{IR}} = \Omega_{\mathcal{C}_{\mathrm{IR}}}$.

5.2 Functoriality and stable corners

Assume the degree-preserving codimension-one restriction of Φ is a strong unitary pivotal tensor functor $F : \mathcal{C}_{\mathrm{UV}} \rightarrow \mathcal{C}_{\mathrm{IR}}$. Write its tensorator as

$$F_2(x, y) : F(x) \otimes F(y) \xrightarrow{\sim} F(x \otimes y).$$

On a representative $f : a \otimes j \rightarrow i \otimes a$, define

$$\mathrm{Ann}(F)[a, f] = [F(a), F_2(i, a)^{-1} F(f) F_2(a, j)]. \quad (5.3)$$

Dinaturality in a makes this assignment descend to the coend. On objects, $\mathrm{Ann}(F)$ sends i to $F(i)$.

Proposition 5.1 (Tube homomorphism). *Equation (5.3) defines a dagger functor $\text{Ann}(F)$. For the chosen source-tagged generators it induces a unital $*$ -homomorphism*

$$\alpha_F : \text{Tub}(\mathcal{C}_{\text{UV}}) \longrightarrow \text{End}_{\text{Ann}(\mathcal{C}_{\text{IR}})}(F\Omega_{\text{UV}}). \quad (5.4)$$

Proof. Naturality of F_2 proves the coend relation. Applying (5.3) before or after gluing two annuli gives the two sides of the tensorator pentagon, so multiplication is preserved. The triangle identity supplies the unit. Pivotality makes the two strand bendings agree, after which unitarity proves dagger preservation. $\square$

Since $F\Omega_{\text{UV}}$ lies in the additive Cauchy closure of Ω_{IR} , choose an isometry

$$v : F\Omega_{\text{UV}} \hookrightarrow \Omega_{\text{IR}}^{\oplus N}, \quad p = vv^\dagger. \quad (5.5)$$

Conjugation by v gives

$$a \longmapsto v \alpha_F(a) v^\dagger \in pM_N(\text{Tub}(\mathcal{C}_{\text{IR}}))p, \quad (5.6)$$

a homomorphism into a corner of a matrix algebra over $\text{Tub}(\mathcal{C}_{\text{IR}})$. Multiplicativity follows from $v^\dagger v = 1$, and the unit in the target corner is p . After a common stabilization, different choices of v give Murray–von Neumann equivalent projections and unitarily conjugate corner maps.

5.3 Tube correspondences and their support

For a represented wall, $\mathcal{K}_R(x, y) \simeq \text{Hom}(Fx, y)$. The UV–IR correspondence comes from the conjoint kernel

$$\mathcal{K}_R^\dagger(y, x) \simeq \text{Hom}(y, Fx).$$

Its generator presentation is

$$E_F = \text{Hom}_{\text{Ann}(\mathcal{C}_{\text{IR}})}(\Omega_{\text{IR}}, F\Omega_{\text{UV}}). \quad (5.7)$$

Set $A = \text{Tub}(\mathcal{C}_{\text{UV}})$ and $B = \text{Tub}(\mathcal{C}_{\text{IR}})$. For $\xi, \eta : \Omega_{\text{IR}} \rightarrow F\Omega_{\text{UV}}$, the actions and inner product are

$$a \cdot \xi \cdot b := \alpha_F(a) \circ \xi \circ b, \quad \langle \xi, \eta \rangle_B := \xi^\dagger \circ \eta.$$

E_F is an A – B correspondence. Such correspondences compose by balanced tensor product after Cauchy completion. Under the isometry v in (5.5), $E_F \simeq pB^N$, hence

$$\mathcal{L}_B(E_F) \cong pM_N(B)p \cong \text{End}_{\text{Ann}(\mathcal{C}_{\text{IR}})}(F\Omega_{\text{UV}}).$$

The left action becomes (5.6) under this identification.

Let $\mathcal{B}$ be a finite semisimple rigid C^* -2-category presenting a weak monoidal Morita context, with $\mathcal{B}_{00} = \mathcal{C}_{\text{UV}}$ and $\mathcal{B}_{11} = \mathcal{C}_{\text{IR}}$. An interface retained as a correspondence has topological sector $\mathcal{M}_R = \mathcal{B}_{01}$, taken to be nonzero and full. Choose simple representatives in each hom category. Its mixed tube corner is

$$E_R := \text{Tub}(\mathcal{B})_{01} = \bigoplus_{\substack{i \in \text{Irr}(\mathcal{C}_{\text{UV}}), a \in \text{Irr}(\mathcal{C}_{\text{IR}}) \\ m \in \text{Irr}(\mathcal{M}_R)}} \text{Hom}_{\mathcal{M}_R}(m \triangleleft a, i \triangleright m). \quad (5.8)$$

Tube multiplication restricts to left and right actions on E_R , and the involution gives

$$\langle x, y \rangle_{\text{IR}} := x^* y \in \text{Tub}(\mathcal{C}_{\text{IR}}).$$

Tube-algebra positivity makes E_R a pre-Hilbert module. Its completion is a $\text{Tub}(\mathcal{C}_{\text{UV}})$ - $\text{Tub}(\mathcal{C}_{\text{IR}})$ correspondence, with associativity determined by the bicategory associators and chosen conjugate solutions [9, Sec. 4.2].

In this weak Morita context, the completed mixed corner is an imprimitivity bimodule [9, Thm. 4.3] with

$$\overline{\text{span}} \langle E_R, E_R \rangle_{\text{Tub}(\mathcal{C}_{\text{IR}})} = \text{Tub}(\mathcal{C}_{\text{IR}}). \quad (5.9)$$

A selected wall sector gives a closed subcorrespondence $E_W \subseteq E_R$ when it is closed under both tube actions. Its detected ideal is

$$I_W = \overline{\text{span}} \langle E_W, E_W \rangle_{\text{Tub}(\mathcal{C}_{\text{IR}})}. \quad (5.10)$$

The quotient $\text{Tub}(\mathcal{C}_{\text{IR}})/I_W$ and the kernel of the left action on E_W are undetected by that sector. Finite semisimplicity makes the displayed norm closures automatic. A nonsemisimple extension requires an annular coend and a specified Hilbert-module completion.

5.4 Tube transport requires tensorator data

For $\mathcal{C} = \text{Vec}_G$, the tube algebra is the convolution algebra of the conjugation groupoid. Its basis $[s \mid j]$ has idempotents $e_j = [1 \mid j]$ satisfying $e_j e_k = \delta_{jk} e_j$. For the trivial functor $\text{Vec}_{\mathbb{Z}_2} \rightarrow \text{Vec}$, sending every labeled tube to 1 gives $0 = e_0 e_1 \mapsto 1$, which violates multiplicativity. The source-tagged object keeps the two summands separate:

$$F\Omega = 1_0 \oplus 1_1.$$

The resulting homomorphism

$$\alpha_F : \mathbb{C}^4 \longrightarrow M_2(\mathbb{C})$$

acts on both tags. A map to $\text{Tub}(\text{Vec}) = \mathbb{C}$ requires a choice of primitive character. The stable corner (5.6), equivalently the correspondence (5.7), carries the canonical transport.

Tube transport also depends on associator data. Two pointed categories with equal fusion rules and distinct associator classes admit an identity-on-labels monoidal functor exactly when a tensorator cochain β solves

$$\delta\beta = \omega_{\text{UV}}/F^* \omega_{\text{IR}}. \quad (5.11)$$

Equality of the two cohomology classes is the obstruction-vanishing condition for a tube map respecting the labeled annular structure. The tensorator supplies transport data beyond labels and fusion coefficients.

Proposition 5.1 applies to degree-preserving codimension-one transmission. The central-extension compositor (4.10) lives one categorical height higher and requires a fusion 2-category and iterated annular trace. An ordinary character records the phase $\chi \circ \gamma$.

6 Connections and curvature

6.1 Differential background maps

Connection data survive transmutation through a smooth map of differential stacks. Gauge descent and curvature fix what that map must preserve.

If the represented kernel is smooth on the connection stack and preserves connective invertible backgrounds, the RG wall induces

$$\Phi^\nabla : \mathbf{BG}_{\mathrm{UV}, \nabla} \longrightarrow \mathbf{BG}_{\mathrm{IR}, \nabla}. \quad (6.1)$$

The map Φ^∇ sends each gauge morphism $\epsilon : \check{A} \rightarrow \check{A}'$ to $\Phi^\nabla(\epsilon) : \Phi^\nabla(\check{A}) \rightarrow \Phi^\nabla(\check{A}')$. Let $\mathbb{D}$ denote the Deligne differential. In an additive abelian cocycle model,

$$\Phi^\nabla(\check{A} + \mathbb{D}\check{\epsilon}) = \Phi^\nabla(\check{A}) + \mathbb{D}\check{\Lambda}_1(\check{A}, \check{\epsilon}). \quad (6.2)$$

The equality represents a specified gauge 2-isomorphism. Compatibility with composition produces $\check{\Lambda}_2$. Higher naturality records gauge-of-gauge transformations. Large gauge transformations enter through the transition cochains and integral Čech data of the connection stack.

For a Lie homomorphism $\varphi : G \rightarrow H$,

$$(P, A) \longmapsto (P \times_\varphi H, d\varphi(A)), \quad F_{\mathrm{IR}} = d\varphi(F_{\mathrm{UV}}). \quad (6.3)$$

6.2 Differential characteristic maps

Choose an integral class $c \in H^{p+2}(BG; \mathbb{Z})$ together with a specified differential refinement [12, Secs. 2–4] [13, Sec. 5.2]:

$$\widehat{c} : \mathbf{BG}_\nabla \longrightarrow \mathbf{B}^{p+1}U(1)_\nabla. \quad (6.4)$$

Let $\mathcal{P}$ be its curvature polynomial, whose Chern–Weil class represents the real image of c , and write

$$I_{\mathrm{top}} : \widehat{H}^{p+2}(-; \mathbb{Z}) \longrightarrow H^{p+2}(-; \mathbb{Z})$$

for the underlying integral class. On a principal G -bundle Q with connection A , the refinement obeys

$$I_{\mathrm{top}}(\widehat{c}(Q, A)) = c(Q), \quad \mathrm{curv}(\widehat{c}(Q, A)) = \mathcal{P}(F_A). \quad (6.5)$$

Large-gauge coherence is part of the chosen refinement. Its torsion component has zero curvature and is carried by the flat refinement.

For the second Chern class,

$$\widehat{c}_2 : \mathbf{BSU}(n)_\nabla \longrightarrow \mathbf{B}^3U(1)_\nabla, \quad \mathrm{curv}(\widehat{c}_2(P, A)) = -\frac{1}{8\pi^2} \mathrm{Tr}_{\mathrm{fund}}(F_A \wedge F_A). \quad (6.6)$$

Equation (6.6) uses an anti-Hermitian connection on the principal $SU(n)$ -bundle P and lands in the stack of $U(1)$ two-gerbes with connection. Locally, the three-form potential is

the Chern–Simons form. Let $\Omega_{\mathbb{Z}}^r(X)$ denote the closed r -forms on X with integral periods. A local gauge transformation $g : U \rightarrow SU(n)$ acts by $A^g = g^{-1}Ag + g^{-1}dg$, and the Chern–Simons potential obeys

$$CS_{c_2}(A^g) - CS_{c_2}(A) = dQ_2(A, g) + g^*\eta_{c_2}, \quad \eta_{c_2} \in \Omega_{\mathbb{Z}}^3(SU(n)). \quad (6.7)$$

The integral-period term is the Deligne descent component of $\widehat{c}_2$. An RG wall turns this differential characteristic map into symmetry transmutation when it realizes the exact IR two-form action and trivializes the source action at its original codimension.

The relation $B_{\text{IR}} = dA_{\text{UV}}$ requires coefficient data. The forms part of differential cohomology fits into the standard exact sequence [12, Secs. 2–4] [13, Sec. 5.2]:

$$0 \longrightarrow \Omega^2(M)/\Omega_{\mathbb{Z}}^2(M) \xrightarrow{a} \widehat{H}^3(M; \mathbb{Z}) \xrightarrow{I_{\text{top}}} H^3(M; \mathbb{Z}) \longrightarrow 0,$$

with

$$I_{\text{top}}(a[B]) = 0, \quad \text{curv}(a[B]) = dB.$$

For a line bundle with connection $\check{A} = (L, A)$, the closed form $F_A/(2\pi)$ has integral periods. Its class in the quotient is zero, so the compact-gerbe operation $a([F_A/(2\pi)])$ is trivial. An independent class in $H^2(M; \mathbb{R}/\mathbb{Z})$ supplies a flat gerbe.

For a discrete finite abelian group K and a coefficient homomorphism $q : \mathbb{Z} \rightarrow K$, define

$$\Phi_{q, \text{flat}} : \mathbf{BU}(1)_{\nabla} \xrightarrow{I_{\text{top}}} \mathbf{BU}(1) \simeq K(\mathbb{Z}, 2) \xrightarrow{B^2q} K(K, 2) \simeq \mathbf{B}^2K_{\text{disc}}. \quad (6.8)$$

Evaluation on M gives

$$\Phi_{q, \text{flat}}(\check{A}) = q_*(c_1(L)) \in H^2(M; K), \quad B_{\text{IR}} = q_*c_1(L) \quad (K = \mathbb{Z}_k). \quad (6.9)$$

The notation $[F_A/(2\pi)]_k$ means reduction of the integral characteristic class. Under the coefficient embedding $\mathbb{Z}_k \hookrightarrow \mathbb{R}/\mathbb{Z}$, such a class determines a flat gerbe of order dividing k , whose surface holonomies are k th roots of unity. The finite-coefficient map depends on $c_1(L)$ and accepts nonflat connections. Equation (6.6) retains curvature. An ordinary real two-form takes values in a noncompact coefficient theory.

6.3 Underlying classes and the flat comparison

Requiring Φ^{∇} to refine Φ^{top} includes the homotopy-commutative square

$$\begin{array}{ccc} \mathbf{BG}_{\text{UV}, \nabla} & \xrightarrow{\Phi^{\nabla}} & \mathbf{BG}_{\text{IR}, \nabla} \\ u_{\text{UV}} \downarrow & & \downarrow u_{\text{IR}} \\ \mathbf{BG}_{\text{UV}} & \xrightarrow{\Phi^{\text{top}}} & \mathbf{BG}_{\text{IR}}. \end{array} \quad (6.10)$$

For $T \in \{\text{UV}, \text{IR}\}$, let $\mathfrak{F}_T$ be its gauge-covariant curvature stack and define

$$\mathbf{BG}_{T, \nabla, \text{flat}} := * \times_{0, \mathfrak{F}_T} \mathbf{BG}_{T, \nabla}.$$

Let $f : \mathfrak{F}_{\text{UV}} \rightarrow \mathfrak{F}_{\text{IR}}$ carry the specified homotopy

$$\text{curv}_{\text{IR}} \circ \Phi^\nabla \simeq f \circ \text{curv}_{\text{UV}}, \quad f(0) = 0.$$

The flat inclusions ι_{UV} and ι_{IR} form the square

$$\begin{array}{ccc} \mathbf{BG}_{\text{UV}, \nabla, \text{flat}} & \xrightarrow{\Phi^{\text{flat}}} & \mathbf{BG}_{\text{IR}, \nabla, \text{flat}} \\ \downarrow \iota_{\text{UV}} & & \downarrow \iota_{\text{IR}} \\ \mathbf{BG}_{\text{UV}, \nabla} & \xrightarrow{\Phi^\nabla} & \mathbf{BG}_{\text{IR}, \nabla}. \end{array} \quad (6.11)$$

A flat connection retains its full holonomy representation, including continuous and torsion holonomies. The vertical maps in (6.10) forget the connection and its holonomy.

For an ordinary Lie group G , holonomy gives an equivalence of groupoids

$$\mathbf{BG}_{\nabla, \text{flat}}(M) \simeq \text{Fun}(\Pi_1(M), BG).$$

Network transport also uses the action on flat defect fibers from Definition 2.1. Subdivision and gauge-compatible isotopy relate its cellular representatives.

7 Anomaly matching

Anomaly matching is naturality of determinant lines under a local Wilsonian pushforward. Let Φ also denote the functor on background bordisms induced by Φ .

The anomaly of a d -dimensional theory T is an invertible field theory

$$\mathcal{A}_T : \text{Bord}_{d+1}^{\mathfrak{B}_T^\nabla} \longrightarrow \text{Pic}(\mathcal{V}), \quad (7.1)$$

with T a relative boundary condition in the invertible-anomaly case [14, Sec. 2.3]. Extended relative theories with noninvertible anomaly field theories are treated in [15, Secs. 3.1–3.4]. We allow noninvertible defect labels in the background stack and use $\mathcal{A}_T$ for the ordinary invertible ‘t Hooft anomaly.

Let Z_T denote the relative theory with anomaly $\mathcal{A}_T$. Wilsonian integration splits the fields into retained modes ℓ and eliminated modes h :

$$e^{-S_{\text{eff}}[\ell; A]} = \int_{\pi^{-1}(\ell)} \mathcal{D}h e^{-S_{\text{UV}}[h, \ell; A]}. \quad (7.2)$$

Take the determinant-line pushforward defined by this integral to be compatible with base change and the symmetric-monoidal gluing structure, including disjoint union and collars. Require the low-energy factorization

$$Z_{\text{eff}} \simeq \Phi^* Z_{\text{IR}} \otimes \mathcal{I}_R$$

on bordisms and background morphisms, where $\mathcal{I}_R$ contains massive SPT, Chern–Simons, and Wess–Zumino terms. A local, base-change-compatible realization of this factorization gives a monoidal natural transformation

$$W : Z_{\text{UV}} \longrightarrow \Phi^* Z_{\text{IR}} \otimes \mathcal{I}_R. \quad (7.3)$$

Write $\mathcal{A}_{\text{rel}}$ for the anomaly of $\mathcal{I}_R$.

Theorem 7.1 (Anomaly transport from a local Wilsonian pushforward). *Let $\mathcal{B}_{\text{tr}}$ be a symmetric-monoidal transmitted background-bordism category equipped with the background functor induced by Φ . Set*

$$\mathcal{A}_0 = \mathcal{A}_{\text{UV}}|_{\mathcal{B}_{\text{tr}}}, \quad \mathcal{A}_1 = (\Phi^* \mathcal{A}_{\text{IR}} \otimes \mathcal{A}_{\text{rel}})|_{\mathcal{B}_{\text{tr}}}.$$

Suppose these are symmetric-monoidal functors to $\text{Line}_{\mathbb{C}}$. Assume the Wilsonian pushforward above is defined on anomaly determinant lines and induces a monoidal natural transformation

$$w : j\mathcal{A}_0 \Longrightarrow j\mathcal{A}_1, \quad j : \text{Line}_{\mathbb{C}} \hookrightarrow \text{Vect}_{\mathbb{C}},$$

whose component at every object is nonzero. Then

$$\mathcal{A}_{\text{UV}}|_{\mathcal{B}_{\text{tr}}} \simeq (\Phi^* \mathcal{A}_{\text{IR}} \otimes \mathcal{A}_{\text{rel}})|_{\mathcal{B}_{\text{tr}}}. \quad (7.4)$$

If $\mathcal{A}_{\text{rel}} \simeq \partial\mathcal{C}_R$, where $\partial\mathcal{C}_R$ is the anomaly theory of a local counterterm, the UV and pulled-back IR anomaly classes agree modulo local counterterms.

Proof. Each w_b is a nonzero map between one-dimensional complex vector spaces, so it is an isomorphism. For a background bordism $u : b \rightarrow b'$, naturality gives

$$w_{b'}^{-1} \mathcal{A}_1(u) = \mathcal{A}_0(u) w_b^{-1}.$$

The inverses form a natural transformation. The monoidal equations for w invert to those for w^{-1} , proving (7.4). $\square$

Constructing a nonvanishing Wilsonian pushforward is a dynamical problem. For a fully extended $\text{Pic}(\mathcal{V})$ -valued anomaly, its lift must be an equivalence at every categorical level.

7.1 Cocycle form on the pointed invertible sector

For a background morphism $u : b \rightarrow b'$ in the transmitted sector, naturality of w reads

$$w_{b'} U_{\text{UV}}(u) = (U_{\text{IR}}(\Phi u) \otimes U_{\text{rel}}(u)) w_b. \quad (7.5)$$

After choosing local cocycle representatives, write the projective multipliers additively,

$$U_T(g)U_T(h) = e^{2\pi i a_T(g,h)} U_T(gh), \quad a_T(g, h) \in \mathbb{R}/\mathbb{Z}.$$

Applying (7.5) twice gives

$$\left[e^{2\pi i (a_{\text{UV}} - \Phi^* a_{\text{IR}} - a_{\text{rel}})(g,h)} - 1 \right] w_b U_{\text{UV}}(gh) = 0. \quad (7.6)$$

Since $w_b U_{\text{UV}}(gh) \neq 0$, equation (7.6) implies

$$a_{\text{UV}} = \Phi^* a_{\text{IR}} + a_{\text{rel}} \quad \text{in } \mathbb{R}/\mathbb{Z}.$$

On higher gauge simplices, the Deligne cocycle equation represents this naturality:

$$\check{\alpha}_{\text{UV}} - \Phi^* \check{\alpha}_{\text{IR}} - \check{\alpha}_{\text{rel}} = \mathbb{D} \check{\lambda}_R. \quad (7.7)$$

A change of representatives shifts $\check{\lambda}_R$ by the corresponding coboundary. For every large-gauge loop γ in the transmitted sector, the intertwiner gives

$$\text{Hol}_{\mathcal{A}_{\text{UV}}}(\gamma) = \text{Hol}_{\Phi^* \mathcal{A}_{\text{IR}} \otimes \mathcal{A}_{\text{rel}}}(\gamma).$$

Failure of (7.7) obstructs anomaly-compatible transmutation.

7.2 Wall-bordism correspondences

When $\mathcal{K}_R$ is retained as a profunctor, let Bord_{d+1}^R be a wall-bordism category equipped, as part of the interface data, with functors

$$q_{\text{UV}} : \text{Bord}_{d+1}^R \rightarrow \text{Bord}_{d+1}^{\mathfrak{B}_{\text{UV}}^\nabla}, \quad q_{\text{IR}} : \text{Bord}_{d+1}^R \rightarrow \text{Bord}_{d+1}^{\mathfrak{B}_{\text{IR}}^\nabla}.$$

If coarse graining on the wall induces a pointwise nonzero monoidal transformation between the pulled-back anomaly-line functors, the line-isomorphism proof gives

$$q_{\text{UV}}^* \mathcal{A}_{\text{UV}} \simeq q_{\text{IR}}^* \mathcal{A}_{\text{IR}} \otimes \mathcal{A}_{\text{rel}} \quad \text{on } \text{Bord}_{d+1}^R. \quad (7.8)$$

On this category, the correspondence itself keeps the UV–IR junction.

7.3 Categorical and tube consequences

Let

$$F : \text{Vec}_{G_{\text{UV}}}^{\omega_{\text{UV}}} \longrightarrow \text{Vec}_{G_{\text{IR}}}^{\omega_{\text{IR}}}$$

be induced on simple objects by $\varphi : G_{\text{UV}} \rightarrow G_{\text{IR}}$, with tensorator cochain β . Its coherence condition is

$$\delta\beta = \omega_{\text{UV}}/\varphi^* \omega_{\text{IR}}. \quad (7.9)$$

Tube multiplication and anomaly matching both require the wall-junction coherence equation (7.9). Its failure obstructs the wall action and the induced bimodule. For a general fusion category, the module-wall coherence diagram replaces this scalar equation. In the mixed tube construction, annular transgression gives E_R the corresponding anomaly-twisted bimodule structure.

7.4 Recovery of the central-extension anomaly

Write n for the QFT spacetime dimension in this example. In the notation of [1, Sec. 6.2.1, Eqs. (6.4)–(6.5)], $n = d + 1$. Evaluate the inflow theory on a closed $(n + 1)$ -manifold Y , and take $B \in Z^2(Y; \mathbb{Z}_2)$ and $C \in Z^{n-1}(Y; \mathbb{Z}_2)$. The IR mixed-anomaly phase is

$$\exp \alpha_{\text{IR}}(B, C) = \exp \left(\pi i \int_Y B \smile C \right). \quad (7.10)$$

Take these mixed terms to exhaust the anomaly on the transmitted flat sector, with $\mathcal{A}_{\text{rel}} \simeq \mathbf{1}$. For $A \in Z^1(Y; \mathbb{Z}_2)$, choose an integral lift $\tilde{A}$ and set

$$\beta(A) = \left(\frac{1}{2} \delta \tilde{A} \bmod 2 \right) \in Z^2(Y; \mathbb{Z}_2).$$

By (4.8), $\Phi_e(A) = [\beta(A)]$ on this flat sector. Define the UV mixed-anomaly phase by

$$\exp \alpha_{\text{UV}}(A, C) := \exp \left(\pi i \int_Y \beta(A) \smile C \right).$$

Substitution into (7.10) gives

$$\exp \alpha_{\text{IR}}(\Phi_e A, C) = \exp \left(\pi i \int_Y \beta(A) \smile C \right) = \exp \alpha_{\text{UV}}(A, C). \quad (7.11)$$

Changing the lift by $\tilde{A}' = \tilde{A} + 2m$ changes $\beta(A)$ by $\delta m \bmod 2$, so the phase on closed Y is lift independent. Equation (7.11) proves the pullback equality for the flat mixed-anomaly class. Promoting it to an anomaly-line isomorphism uses the pointwise nonzero determinant-line transformation from Theorem 7.1. The UV data test the pullback along Φ_e . Values on the remaining IR backgrounds are independent data [1, Sec. 6.4].

8 Model-level construction

For a concrete RG flow, construct the coefficient fibration and colored operad map from its field and defect data. Equation (2.5) shows what is missing when connection and defect factors remain independent: the fibration must encode their coupling. Represent the resulting physical wall kernel on the transmitted sector with the locality and gluing coherences from Section 3. A compatible section of the IR wall action proves right-principality and fixes Φ by enriched Yoneda.

Anomaly transport adds the local determinant-line pushforward from Section 7. Pointwise nonvanishing turns its monoidal map into the anomaly-line isomorphism. For each flow, the wall section and this pushforward decide the lift from correspondence to transmission map.

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